COMPUTER PROGRAMMING

LAB REPORT 2

TITLE

Basic Syntax and Input/Output in C++

1. OBJECTIVE

- The objective of this lab is to understand the basic structure of a C++ program. This lab explains how to use input and output statements, work with different data types, apply proper indentation, and perform calculations using arithmetic operators.

2. THEORY

- A C++ program follows a standard structure that includes header files, the main() function, and program statements. The iostream library is commonly used to handle input and output operations.

- The cout statement is used to display output on the screen, while cin is used to receive input from the user. The symbols << and >> are called stream insertion and stream extraction operators.

- C++ is a statically typed programming language, meaning every variable must be declared with a specific data type before use.

  • int – used for integer numbers
  • double – used for decimal numbers
  • char – used for single characters
  • bool – used for true or false values
  • string – used for text

3. SOFTWARE USED

• DEV-C++ IDE
• C++ Compiler (included with DEV-C++)

PROGRAM 1

Hello World and User Greeting

1. Start the program.
2. Declare a variable to store the user's name.
3. Ask the user to enter their name.
4. Display a greeting message.
5. End the program.

```
#include <iostream>
using namespace std;

int main()
{
    string name;

    cout << "Enter your name: ";
    cin >> name;

    cout << "Hello " << name
         << "! Welcome to C++ programming.";

    return 0;
}
```

PROGRAM 2

Simple Calculator Using Arithmetic Operators

6. Start the program.
7. Declare two variables for numbers.
8. Take input values from the user.
9. Perform addition, subtraction, multiplication, and division.
10. Display the calculated results.
11. End the program.

```
#include <iostream>
using namespace std;

int main()
{
    int a, b;

    cout << "Enter two numbers: ";
    cin >> a >> b;

    cout << "Addition = " << a + b << endl;
```

```
    cout << "Subtraction = " << a - b << endl;
    cout << "Multiplication = " << a * b << endl;
    cout << "Division = " << a / b << endl;

    return 0;
}
```

CONCLUSION

In this lab, we learned the basic structure of a C++ program. We practiced using input and output statements, worked with different data types, and performed calculations using arithmetic operators. This lab provided a strong foundation for understanding basic C++ programming concepts.